1. Markov Networks: ..... *Total: 14 marks*

(1) Consider a small group of friends: Alice, Bob, Carol, Dave, and Emma. They regularly influence each other's decisions about attending social events, and their influence is symmetric: if one person influences another, the reverse also holds. However, not all pairs of friends directly influence each other, and their relationships form a specific pattern of dependence and independence.

Relationships:

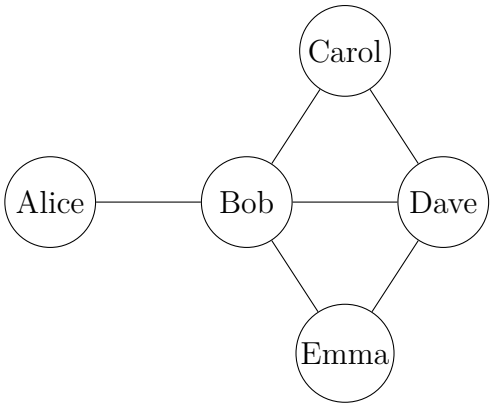
- Alice and Bob frequently discuss their plans, so they strongly influence each other's decisions.
- Bob, Carol, and Dave form a close trio, where each member influences the other two.
- Emma is not directly influenced by Alice, but she often relies on Bob and Dave for decision-making.

No other pairs of friends influence each other directly. Model each friend's decision as a binary random variable (1 if the person attends the event, 0 otherwise); for brevity you may abbreviate the variables as A, B, C, D, E.

Tasks: Design a Markov network to represent the influence relationships in this group of friends.

(5 marks) (a). Draw the Markov network showing the dependencies between Alice, Bob, Carol, Dave, and Emma based on the relationships described above.

Solution:



An undirected graph with one node per friend and the six edges Alice–Bob; Bob–Carol, Bob–Dave, Carol–Dave (the trio is a triangle); Bob–Emma, Dave–Emma. There is no edge between Alice and Carol, Dave or Emma, and none between Carol and Emma. Any layout with exactly this edge set receives full marks.

Marking: 1 mark for an undirected graph with the five nodes; 1 mark each for the Alice–Bob edge, the Bob–Carol–Dave triangle, and the two Emma edges; 1 mark for including no spurious edges. Deduct 1 mark for directed edges.

(4 marks) (b). Using the network, identify at least two conditional independencies and explain why they are independent.

Solution: In an undirected graph, conditional independence is read off by graph separation: $X \perp Y \mid Z$ holds for every distribution that factorises over the graph whenever every path between X and Y passes through a node in Z (i.e. Z blocks all paths). Any two of the following statements, each with a correct justification, earn full marks.

- $A \perp \{C, D, E\} \mid B$. Bob is Alice's only neighbour, so $\{B\}$ is Alice's Markov blanket and every path from Alice to another node passes through Bob. Observing Bob blocks all of them. Any sub-statement, such as $A \perp D \mid B$ or $A \perp E \mid B$, follows from this one by the decomposition property and is also acceptable.
- $C \perp \{A, E\} \mid \{B, D\}$. Carol's neighbours are Bob and Dave, so $\{B, D\}$ is Carol's Markov blanket and separates her from Alice and Emma.
- $E \perp \{A, C\} \mid \{B, D\}$. Symmetric to the previous case: Emma's Markov blanket is $\{B, D\}$.

The graph implies no marginal (unconditional) independencies, since it is connected. In particular, $A \perp E$ is *not* implied, even though Alice and Emma do not influence each other directly: the dependence flows through Bob. A student who lists $A \perp E$ without a conditioning set has misread the question.

Marking: 1 mark for each correct independence statement (up to 2) and 1 mark for each correct separation-based justification (up to 2). An independence statement that is false under the graph (e.g. $C \perp E \mid B$, which fails because of the path Carol–Dave–Emma) receives 0 for that item.

(5 marks) (c). Design the factors, write the joint distribution of all variables and factorize them based on your design. Explain the underlying principles, why you design such factors and why such a factorization is appropriate.

Solution: Principles. The maximal cliques of the graph are $\{A, B\}$, $\{B, C, D\}$ and $\{B, D, E\}$. A Gibbs distribution factorises over the graph

provided every factor's scope is a clique (a complete subgraph); this is what guarantees that the distribution satisfies the separation-based independencies identified in (b). Conversely, by the Hammersley–Clifford theorem, every strictly positive distribution that satisfies those independencies can be written as a product of factors over the maximal cliques. Including one factor per *maximal* clique therefore gives the most general parameterisation. Factors over proper subsets of a clique (pairwise or unary) are permitted but add no expressive power, since each can be absorbed into a factor over a maximal clique containing it. Dropping a maximal clique (e.g. using only pairwise factors) still yields a valid Gibbs distribution over the graph, but one that cannot express three-way interactions within the trio or within Bob–Dave–Emma, so it would be a weaker answer.

Design 1 (one factor per maximal clique):

$$P(A,B,C,D,E) = \frac{1}{Z}\,\phi_1(A,B)\,\phi_2(B,C,D)\,\phi_3(B,D,E),$$
$$Z = \sum_{a,b,c,d,e} \phi_1(a,b)\,\phi_2(b,c,d)\,\phi_3(b,d,e).$$

Each ϕ_i is a non-negative table over the 2^k joint assignments of its k variables, and Z is the partition function that normalises the product to a probability distribution. For example, ϕ_1 encodes the strong Alice–Bob influence by assigning large values to the agreeing assignments $(A,B) = (1,1)$ and $(0,0)$ and small values to $(1,0)$ and $(0,1)$; ϕ_2 and ϕ_3 can likewise favour assignments in which the three members agree.

Design 2 (adding sub-clique factors):

$$P(A,B,C,D,E) = \frac{1}{Z}\,\phi_1(A,B)\,\phi_2(B,C,D)\,\phi_3(B,D,E)\,\phi_4(B,C) \times \phi_5(A)\,\phi_6(C)\,\phi_7(D).$$

The unary factors can be read as each friend's baseline tendency to attend and the pairwise factor ϕ_4 as the direct Bob–Carol influence; such a design is more interpretable but represents exactly the same family of distributions as Design 1.

Why the factorisation is appropriate. Every factor scope is a clique, so the distribution is Markov with respect to the graph and entails exactly the independencies in (b); and every maximal clique carries a factor, so no interaction permitted by the graph is ruled out.

Marking: 1 mark for identifying the three maximal cliques; 2 marks for a correct normalised joint distribution in which every factor scope is a clique and every maximal clique is covered (1 mark if a maximal clique is missing); 2 marks for the explanation (clique scopes, Hammersley–Clifford

or an equivalent argument, role of Z). A factor whose scope is not a clique, such as $\phi(A, E)$ or $\phi(C, E)$, is incorrect and loses the 2 distribution marks.